

Introduction to Power BI

A foundational training handout covering what Power BI is, its components, workflow, and core building blocks.

1. What is Power BI?

Power BI is a **business intelligence and data visualization platform** developed by Microsoft. It lets users connect to a wide range of data sources, transform and model that data, and turn it into interactive reports and dashboards that support data-driven decision-making.

- Used by organizations of every size — from individual analysts to Fortune 500 enterprises.
- Combines self-service data preparation, modeling, visualization, and sharing in one connected toolset.
- Part of the broader Microsoft Power Platform (alongside Power Apps, Power Automate, and Power Pages).

2. Why Organizations Use Power BI

- **Ease of use** — drag-and-drop report building, no coding required for most tasks.
- **Broad connectivity** — connects to Excel, SQL databases, cloud services, APIs, and hundreds of other sources.
- **Powerful modeling** — build relationships between tables and write calculations using DAX.
- **Interactive visuals** — charts, maps, and tables that respond to clicks, filters, and slicers.
- **Easy sharing** — publish reports to the web/organization and view them on desktop or mobile.
- **Cost-effective** — a free Desktop app, with affordable licensing for collaboration and sharing.
- **Integrates with the Microsoft ecosystem** — Excel, Teams, SharePoint, Azure, and Dynamics 365.

3. Core Components of Power BI

Component	What It Is	Used For
Power BI Desktop	Free Windows application	Connecting to data, building models, designing reports
Power BI Service	Cloud platform (app.powerbi.com)	Publishing, sharing, collaborating, scheduling refreshes
Power BI Mobile	iOS / Android / Windows app	Viewing reports and dashboards on the go
Power BI Report Builder	Paginated report designer	Pixel-perfect, printable reports (invoices, statements)
Power BI Gateway	On-premises data connector	Refreshing cloud reports from on-premises data sources
Power BI Embedded	Developer API/service	Embedding reports into custom applications

Most learning and report-building happens in **Power BI Desktop**; the finished report is then published to the **Power BI Service** for sharing and collaboration.

4. The Typical Power BI Workflow

- **1. Connect** — Get Data from a source (Excel, SQL, web, cloud service, etc.).
- **2. Transform (ETL)** — Clean and reshape data in Power Query Editor.
- **3. Model** — Build relationships between tables and write DAX measures/columns.
- **4. Visualize** — Design report pages using charts, tables, maps, cards, and slicers.
- **5. Publish** — Push the report to the Power BI Service.
- **6. Share & Collaborate** — Share dashboards, set up apps/workspaces, schedule refreshes.

This mirrors the classic ETL (Extract, Transform, Load) process, with visualization and sharing layered on top.

5. Power BI Desktop Interface Overview

Area	Description
Ribbon (top)	Home, Insert, Modeling, View, Help tabs — main commands and actions
Report view	Default canvas where visuals are placed and arranged on report pages
Table/Data view	Shows the raw data in each loaded table, row by row
Model view	Shows tables and the relationships between them (diagram view)
Visualizations pane	Gallery of chart/visual types plus formatting options for the selected visual
Fields pane	Lists all loaded tables and their columns/measures, used to build visuals
Filters pane	Apply filters at the visual, page, or report level

6. Key Building Blocks

- **Tables** — data loaded from sources, shown as rows and columns (like a spreadsheet).
- **Columns** — fields within a table; can be existing data or calculated columns.
- **Measures** — DAX calculations that aggregate data dynamically based on report context (e.g. Total Sales).
- **Relationships** — links between tables (typically one-to-many) that let filters flow across the model.
- **Visuals** — charts, tables, cards, maps, and other elements used to display data on a report page.
- **Dashboards** — single-page collections of pinned visuals from one or more reports (Service only).
- **Workspaces** — collaborative spaces in the Power BI Service where reports and dashboards are organized and shared.

7. The Data Model: Star Schema

Power BI models typically follow a **Star Schema** — one or more central **Fact tables** (transactional data like Sales) surrounded by **Dimension tables** (descriptive data like Customer, Product, Date) that the fact table relates to.

- **Fact table** — contains measurable, numeric data (Amount, Quantity, Profit) and foreign keys.
- **Dimension table** — contains descriptive attributes (Customer Name, Product Category, Date).
- A well-built Date dimension table is essential for time intelligence calculations.
- Star schema keeps the model simple, fast, and easy to write DAX against — the recommended approach over flat, single-table datasets.

8. Power BI Licensing Overview

Tier	Who It's For	Key Notes
Power BI Desktop	Anyone building reports	Free to download and use
Power BI Free	Individual use	Publish to "My Workspace", limited sharing
Power BI Pro	Individual users/teams sharing reports	Per-user monthly license, needed to share/collaborate
Power BI Premium	Larger organizations	Bigger datasets, more frequent refresh, advanced AI features (Per User or Capacity-based)
Power BI Embedded	Developers/ISVs	Embed reports in custom applications, capacity-based pricing

Note: Exact plans, pricing, and feature limits change periodically — always check Microsoft's official Power BI pricing page for current details before advising students.

9. Common Use Cases

- Sales and revenue performance dashboards
- Financial reporting and budget vs. actual tracking
- HR analytics (headcount, attrition, hiring pipeline)
- Marketing campaign performance and ROI tracking
- Operations and supply chain monitoring
- Customer service and support ticket analysis
- Executive summary dashboards combining data across departments

10. Getting Started Checklist

- Download and install **Power BI Desktop** (free from Microsoft).
- Practice connecting to a simple source first — an Excel file or CSV.
- Learn **Power Query** basics for cleaning data before it loads into the model.
- Build a simple two-table model (e.g. Sales + Date) and create one relationship.
- Write a few basic **DAX measures** (SUM, AVERAGE, COUNT) before moving to CALCULATE and time intelligence.
- Design a first report page with 2–3 visuals and a slicer.
- Publish to the Power BI Service and explore sharing options.